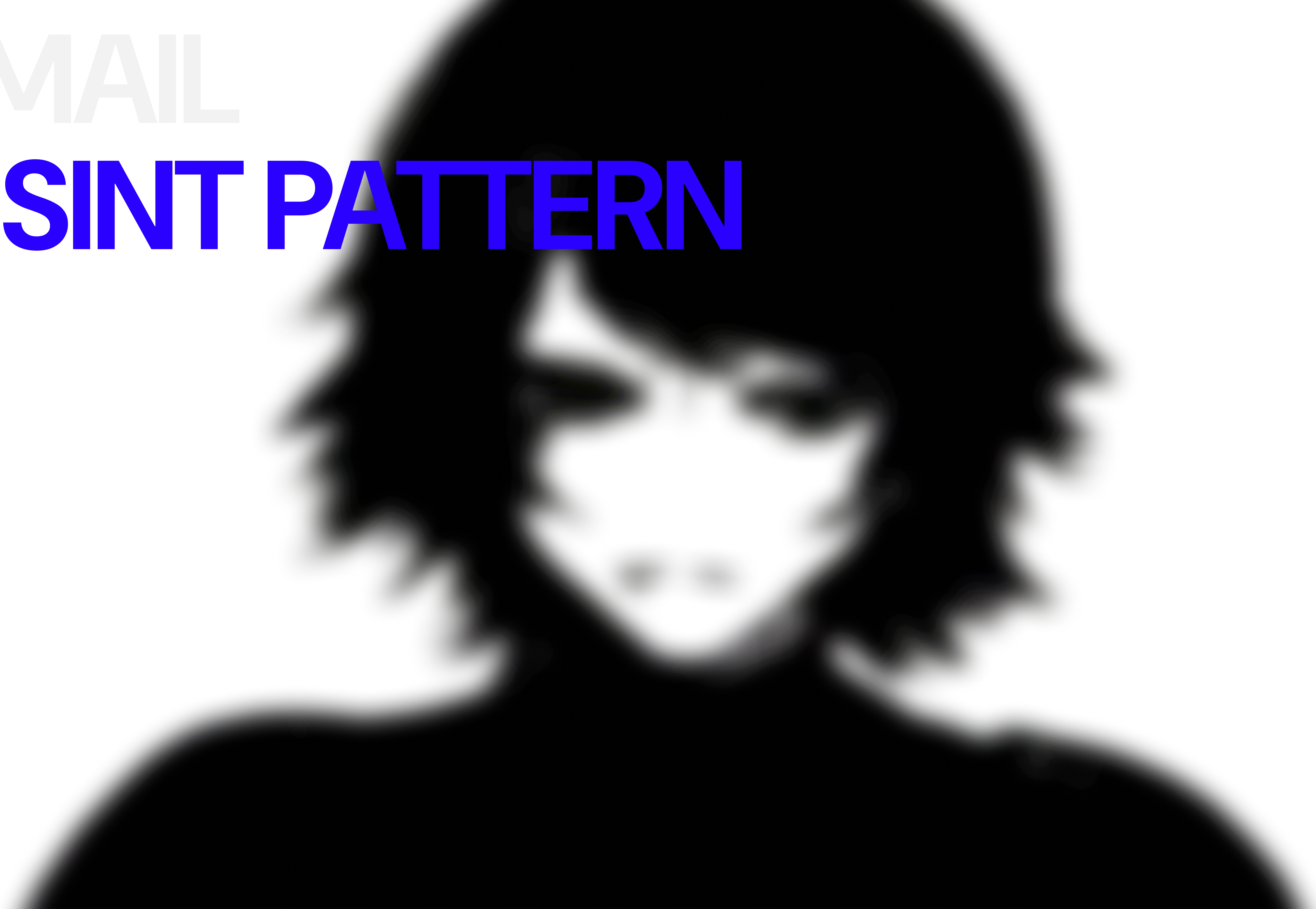
# EMAIL
# OSINT PATTERN



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# Introduction

In OSINT investigations, an email address is often the most valuable starting point. Unlike usernames or real names, emails are deeply connected to online services, leaked databases, domains, social accounts, and other digital traces linked to an individual or organization.

# SMTP — Verifying Whether an Email Address Exists

**SMTP verification** is a basic yet highly effective method for validating email addresses. Instead of sending a real email, it communicates directly with the mail server to determine whether a mailbox exists.

In OSINT, this serves as an **initial filtering step** — helping identify whether an address is **valid, fake, or worth further investigation.**

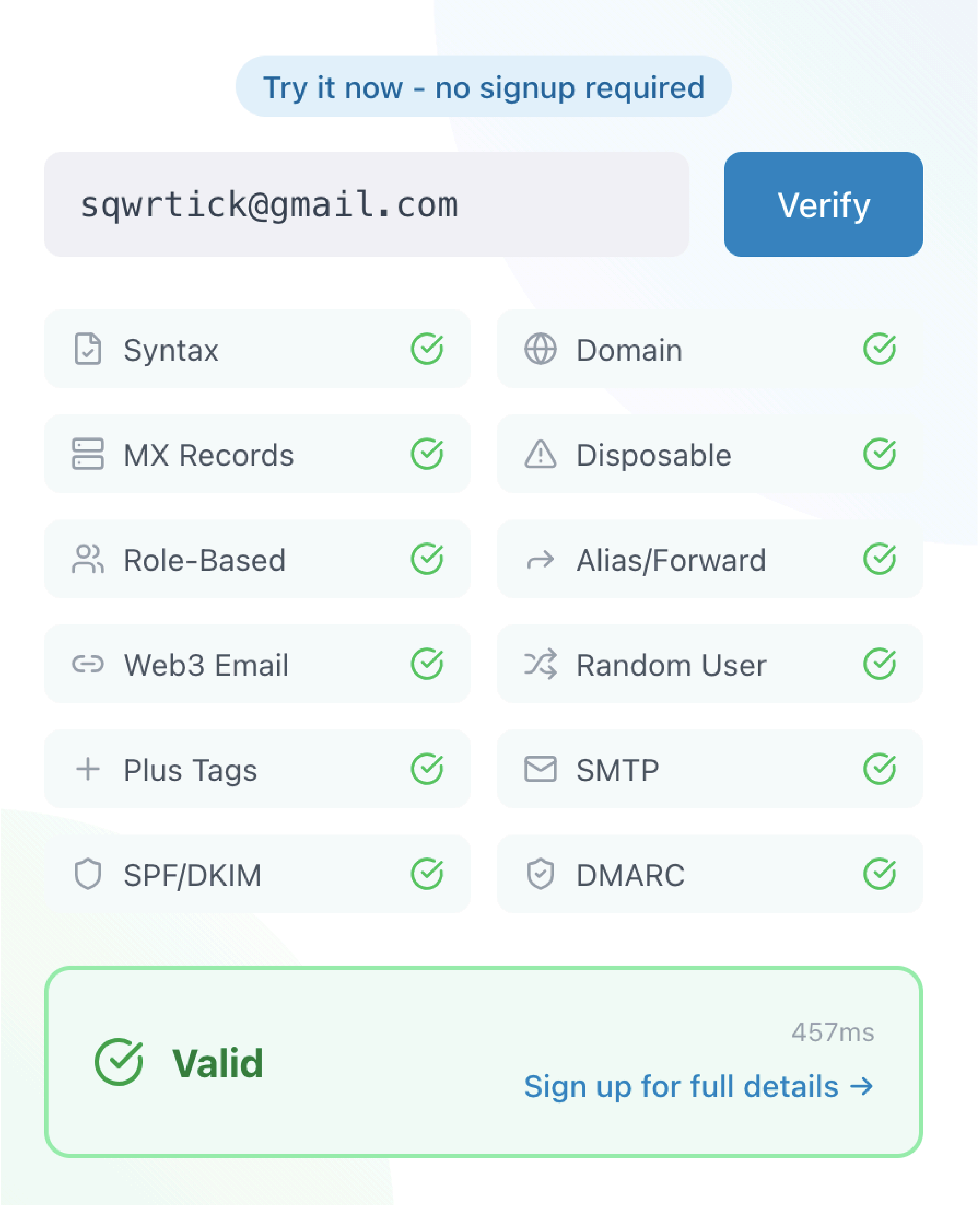
## How it works:

- identify the domain's MX record
- establish a connection to the SMTP server
- submit the target email address
- analyze the server response to determine mailbox validity

## Tools for SMTP verification:

- https://emailvalidator.io/ - simple web interface.

## Example of SMTP verification results:



# Epios — Fast Email Reconnaissance

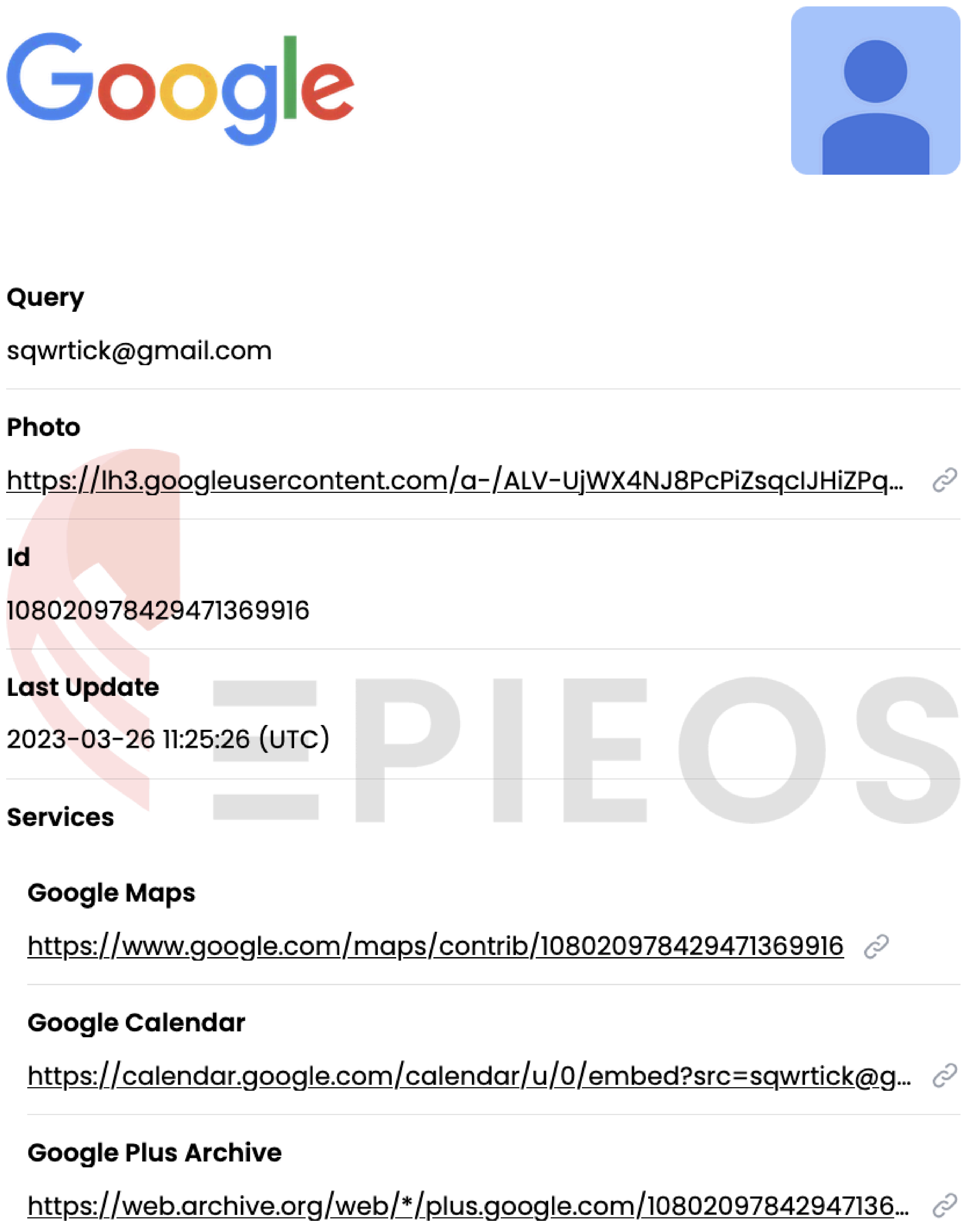
**Epios** is an **OSINT platform** that aggregates publicly available data and helps investigators identify information linked to **email addresses**, phone numbers, and other digital identifiers.

In email-based investigations, it is primarily used for **Gmail account analysis**.

## What Epios can reveal from an email address:

- publicly exposed identity information
- Google profile image and account metadata
- linked Google ecosystem services
- Google Maps, Calendar, and archived Google+ traces
- unique Google account identifiers
- associated platforms and external resources

## Example of the output:



# Checking Which Services Use an Email Address

After validating an email address, the next step is identifying which **platforms and services** are linked to it. Since email addresses are often used as **primary login identifiers**, they can help uncover a person's **digital footprint** across multiple online services.

## Tools designed for this type of reconnaissance rely on several different techniques:

- testing password recovery mechanisms
- analyzing publicly accessible service endpoints
- identifying known response patterns from websites
- correlating data from open sources

## Popular tools for account association checks:

- Blackbird GitHub — a CLI-based OSINT tool for searching accounts using email addresses, usernames, and other identifiers.
- WhatsMyName — a project with a continuously updated database of platforms and detection signatures.

# Example of Blackbird output:


```
[
    {
        "name": "Spotify",
        "url": "https://spclient.wg.spotify.com/signup/public/v1/account?
validate=1&email=german@gmail.com",
        "category": "music",
        "status": "FOUND",
        "metadata": null
    },
    {
        "name": "Chess.com",
        "url": "https://www.chess.com/callback/email/available?email=german@gmail.com",
        "category": "gaming",
        "status": "FOUND",
        "metadata": null
    },
    {
        "name": "Adobe",
        "url": "https://auth.services.adobe.com/signin/v2/users/accounts",
        "category": "misc",
        "status": "FOUND",
        "metadata": [
            {
                "schema": "JSON",
                "type": "Image",
                "name": "Avatar",
                "path": [
                    0,
                    "images",
                    "230"
                ],
                "downloaded": false,
                "value": "https://pps.services.adobe.com/api/profile/image/default/bb99dbb8-eab4-
438d-ab41-f2f9f1b8ead3/230"
            }
        ]
    },
    {
        "name": "El Mundo",
        "url": "https://seguro.elmundo.es/ueregistro/v2/usuarios/registro",
        "category": "shopping",
        "status": "FOUND",
        "metadata": null
    },
    {
        "name": "Eventbrite",
        "url": "https://www.eventbrite.com/api/v3/users/lookup/",
        "category": "social",
        "status": "FOUND",
        "metadata": [
            {
                "schema": "JSON",
                "type": "String",
                "name": "User ID",
                "path": [
                    "user_id"
                ],
                "value": "193341902419"
            }
        ]
    },
    {
        "name": "Duolingo",
        "url": "https://www.duolingo.com/2017-06-30/users?email=german@gmail.com",
        "category": "hobby",
        "status": "FOUND",
        "metadata": null
    }
]
```

# Data Breaches — Checking Whether an Email Address Has Been Compromised

A key part of **email OSINT** is checking whether the address has appeared in **public data breaches** or **leaked databases**.

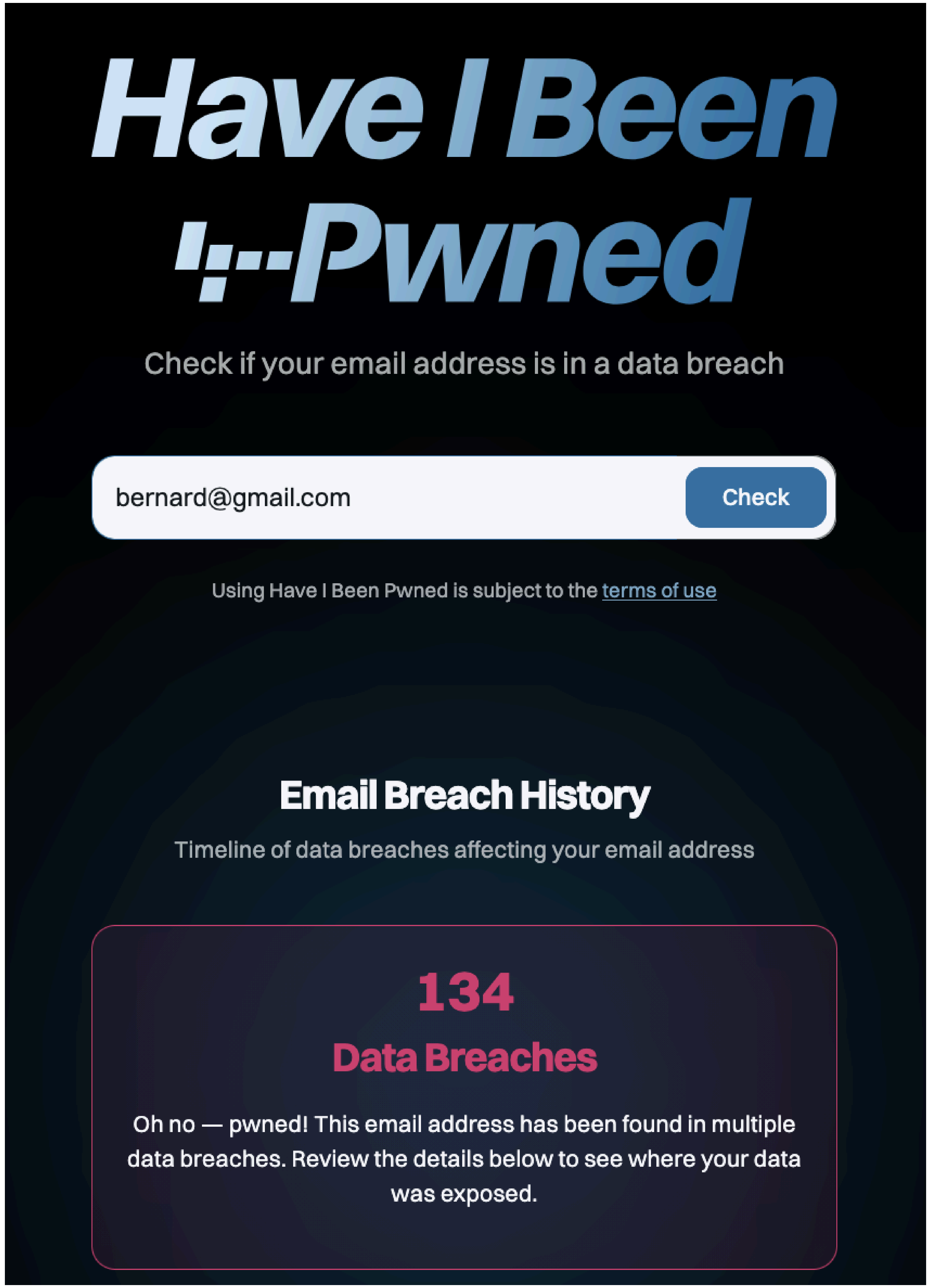
## Information commonly found in breaches:

- email addresses
- usernames or nicknames
- hashed or even plaintext passwords
- account registration dates
- IP addresses
- phone numbers

## Tools for breach analysis:

- Have I Been Pwned — checks whether an email appears in known breaches.
- Intelligence X — platform for searching leaked and indexed data.
- Leak-Lookup — breach-focused email and credential search service.

## Example output from Have I Been Pwned:



# Google Dorking — Finding Public Mentions of an Email Address

**Google Dorking** is the use of **advanced search operators** to locate publicly accessible mentions of an **email address** across the internet — including websites, forums, documents, paste services, and indexed files.

**With properly crafted queries, it is often possible to discover:**

- forum posts and comment sections
- paste-service publications
- leaked documents and exposed files
- public profiles and contact pages
- resumes, PDFs, DOCX files, and spreadsheets
- technical logs and configuration files

## Basic Google dorks for email investigations

- **Exact match search**
  "example@gmail.com"
- **Search on specific websites**
  site:github.com "example@gmail.com"
  site:vk.com "example@gmail.com"
  site:pastebin.com "example@gmail.com"
- **Search by file type**
  filetype:pdf "example@gmail.com"
  filetype:doc "example@gmail.com"
  filetype:xls "example@gmail.com"
- **Search open directories and listings**
  intitle:"index of" "example@gmail.com"
- **Search logs and configuration files**
  "example@gmail.com" "password"
  "example@gmail.com" "login"

# Password Recovery Enumeration — Phone Numbers and Backup Emails

Another useful OSINT technique is **analyzing account recovery mechanisms.** Services like **Google** may reveal **partially masked recovery data** during the **password reset process.**

### This can sometimes expose:

- linked phone numbers
- country codes
- partially masked digits
- recovery email addresses
- masked backup emails (e.g. ***@outlook.com)

Even limited information can help correlate identities, identify geographic links, and support further investigation.

### This method is commonly used to:

- verify a target's country
- correlate phone numbers with leaks or social media
- gather additional reconnaissance data

# OPSEC Notes

**Recovery requests** are typically **logged by the provider,** including **IP address, region,** and **browser data.** Frequent requests may also trigger **CAPTCHAs** or **temporary restrictions.**

### For better OPSEC:

- use proxies or Tor
- avoid personal accounts and primary IPs
- do not perform large-scale checks from one source

In email OSINT, **OPSEC is as important as the data itself.**

# Conclusion

An email address is far more than just a communication method — it is one of the most **stable and informative identifiers** in the digital world. Even with minimal starting data, it can be used to uncover **services**, **breaches**, **public mentions**, and other traces that gradually build a **broader profile** of a target.

No single technique is effective on its own. The real value comes from **correlating data across multiple sources: SMTP checks** validate the address, **breaches** reveal historical exposure, account associations map connected services, while **Google Dorking**, **Epios**, **recovery emails**, and **phone numbers** provide additional context and **investigative pivots**.

At the same time, **OPSEC** should never be overlooked. Many of these techniques leave traces if used carelessly. **Proxies**, **Tor**, and **separation between personal and research accounts** should be considered a basic requirement for any serious **OSINT workflow**.

The end.